

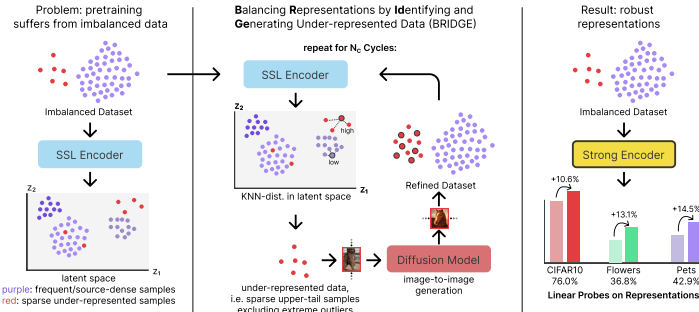


Figure 1: Overview of BRIDGE. Pretraining, sparsity scoring in the current representation and targeted image-to-image generation alternate over cycles. The budget is spent on a mode-centered band inside the upper tail of the score distribution, which excludes the extreme tail.

BALANCING REPRESENTATIONS BY IDENTIFYING AND GENERATING UNDERREPRESENTED DATA

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ABSTRACT

Self-supervised learning inherits the coverage of its source data. When an unlabeled corpus is long-tailed, the learned representation becomes dense around frequent modes and thin where coverage is weak, and a change of training objective adds none of the missing support. BRIDGE is a label-free feedback loop in which the current encoder scores every source image by mean k NN distance, a budgeted acquisition rule selects a sparse but recoverable band of that score distribution, a frozen image-to-image diffusion model adds local variants of the selected anchors, and pretraining continues on the enlarged source set. We formulate acquisition as recoverability-weighted allocation of local support and derive conditions under which a finite budget should favor sparse interior regions and be reallocated after every repair. Across five long-tailed and web-sourced regimes, BRIDGE gives the best seven-task linear-probe and k NN transfer, and paired full-schedule experiments extend the gain to ViT-S under contrastive, self-distillation and masked-reconstruction objectives, to low-label fine-tuning and to semantic segmentation. Matched controls cover uniform and extreme-tail acquisition, conventional augmentation, oracle restoration of withheld real images, language-mediated generation and direct distillation of the generator’s features. The acquisition rule has no measurable effect after a single repair stage, where nine score bands lie within 1.32 points, and the same rules separate by 8.0 points over five stages. The benefit is produced by repeated re-estimation of where support is missing.

1 INTRODUCTION

Self-supervised learning is the standard route to transferable visual representations from unlabeled images, through contrastive learning, self-distillation and masked prediction (Chen et al., 2020; He et al., 2020; Caron et al., 2021; He et al., 2022; Oquab et al., 2024; Siméoni et al., 2025). All of them learn from whatever the source corpus contains. When that corpus is long-tailed, the representation concentrates around frequent modes, covers rare regions weakly and transfers worse (Tian et al., 2021; Assran et al., 2023; Kukleva et al., 2023; Jiang et al., 2021). Prior work has responded on the model side, through a temperature schedule over contrastive training (Kukleva et al., 2023),

a pruned self-competitor that emphasizes rare samples (Jiang et al., 2021) or imported external out-of-distribution data (Bai et al., 2023), and all of it leaves the source set fixed.

We ask whether the source set can be repaired while learning proceeds. After each pretraining stage, BRIDGE embeds the current source set with the current encoder, scores every image by its mean distance to its k nearest neighbors, selects a sparse but semantically coherent band of that distribution, and generates local variants of the selected images with a frozen image-to-image diffusion model, after which pretraining continues on the enlarged set (Figure 1). The encoder identifies where its own representation is thin and the generator adds support there. No labels, class frequencies or downstream signals enter the loop.

Two decisions determine whether such a loop helps, where a finite generation budget is spent and whether the allocation is re-estimated after each repair. The largest scores mark a tail populated by artifacts, degenerate crops and images the generator cannot vary faithfully, while uniform sampling spends most of the budget in regions that are already well supported. BRIDGE restricts acquisition to the 75th to 99th percentile band of the score distribution, centers a candidate pool on the mode of that band and covers the pool with farthest-point sampling. We formalize both decisions as budgeted allocation over regions of representation space with diminishing returns to local support and reparability that falls in the extreme tail. The optimal band is then interior, greedy reallocation after each intervention is optimal, and any rule applied once has an effect bounded by the ratio of budget to existing support.

The last statement describes the experiments. When every score band from the densest quartile to the extreme tail receives the same budget after a single repair stage, the nine seven-task averages lie within 1.32 points, the run-to-run resolution of the pipeline, and the same rules separate by 8.0 points once the loop runs for five stages with re-estimation between them. The contribution of BRIDGE lies in the feedback, and none of its components produces the effect alone.

- We formulate label-free data repair as recoverability-weighted allocation of local support and prove that sparse interior regions carry the largest marginal value, that greedy reallocation after each repair is optimal under diminishing returns, and that any single-stage allocation has an effect bounded by the ratio of budget to existing support.
- We instantiate this policy in BRIDGE, where mean k NN distance estimates local support, a trimmed mode window selects a recoverable sparse band, farthest-point sampling limits redundancy and a frozen image-to-image generator supplies variants.
- We evaluate five source regimes and seven transfer tasks under linear-probe and k NN evaluation, with paired ViT-S experiments across four SSL objectives, low-label fine-tuning, semantic segmentation and matched controls for uniform, extreme-tail, one-shot, conventional, oracle, language-mediated and distillation alternatives.
- We separate the acquisition rule from the loop by running identical rules under one repair stage and under five, and locate the effect in the repeated re-estimation.

2 RELATED WORK

Self-supervised learning under skewed sources. Contrastive learning, momentum contrast, self-distillation and masked reconstruction have made unlabeled pretraining routine (Chen et al., 2020; He et al., 2020; Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025; He et al., 2022), and their sensitivity to source imbalance is established. Divide and Contrast recovers performance on uncured data by clustering and specializing experts (Tian et al., 2021), the hidden uniform cluster prior explains why standard objectives favor balanced sources (Assran et al., 2023), and frequency-aware learning reweights by estimated class frequency (Lin et al., 2023). The methods closest to our comparison change the objective on a fixed corpus. SDCLR prunes a self-competitor so that persistently difficult samples receive more gradient (Jiang et al., 2021), Temperature Schedules emphasize head and tail at different times (Kukleva et al., 2023), and COLT draws external out-of-distribution images into the batch (Bai et al., 2023). BRIDGE holds the objective fixed and changes the source set, so it is complementary, and we report the combination with TS and SDCLR.

Density, rarity and representation-driven curation. Mean deep nearest-neighbor distance is an effective nonparametric out-of-distribution score in high-dimensional feature spaces (Sun et al., 2022),

and we use it as a local-support estimate on the training set itself. Data-centric work uses representations to decide what to keep. Pruning by prototype distance beats power-law scaling (Sorscher et al., 2022), concept-cluster complexity prunes web-scale corpora (Abbas et al., 2024), and data quality is studied in its own right (Whang et al., 2023). These methods remove examples. SMOTE adds them by interpolating labeled minority examples (Chawla et al., 2002), and DOPING oversamples the low-density edge of a learned latent distribution for unsupervised anomaly detection (Lim et al., 2018). DOPING is the closest antecedent to acquiring by latent rarity. It targets the extreme edge, uses no image generator and does not recompute its target after the representation changes, and our top-tail control is a DOPING-style rule under our generator and budget.

Budgeted acquisition and coverage. Active learning is the formal precedent for spending a finite acquisition budget under a learned representation. The core-set formulation treats acquisition as k -center coverage and selects with the greedy farthest-first traversal we use inside the mode window, which carries a factor-two approximation guarantee (Sener and Savarese, 2018; Gonzalez, 1985), and ProbCover maximizes covered probability mass at a fixed radius (Yehuda et al., 2022). Hachohen et al. (2022) show that the preferred strategy reverses with budget, dense regions being better targets when the budget is small relative to the corpus, which our single-stage sweep reproduces. Active learning acquires labels for existing images, whereas BRIDGE creates images without labels, and its scoring function is the encoder being trained, so each acquisition changes the function that makes the next one.

Generative augmentation for supervised recognition. Denoising diffusion and its latent-space variants (Ho et al., 2020; Rombach et al., 2022) make image-conditioned editing possible through SDEdit, which adds noise to an image and denoises it under the model’s prior (Meng et al., 2022), and that operator is our repair step. DA-Fusion applies it to few-shot supervised recognition with class prompts and domain adaptation (Trabucco et al., 2024). Class-conditional generation at ImageNet scale improves supervised accuracy (Azizi et al., 2023), and the utility of synthetic data depends on prompt design, filtering, diversity, guidance and the domain gap to the target (He et al., 2023; Fan et al., 2024). TADA, the closest targeted method, identifies examples a supervised classifier learns slowly, generates faithful variants for them, and reports that augmenting a targeted third of the data beats augmenting all of it (Nguyen et al., 2026). Every method in this family uses class identity or a supervised loss to decide what to augment, whereas BRIDGE uses neither, and we adapt the TADA criterion to a label-free learning-speed signal as a direct baseline.

Synthetic data for representation learning. Synthetic imagery has served as a training source for self-supervised features since Ren and Lee (2018). Representations can be learned from procedural noise and image programs (Baradad et al., 2021; 2022), from the latent neighborhoods of a black-box generator (Jahani et al., 2022), from synthetic ImageNet clones (Sariyildiz et al., 2023) and from text-to-image samples used as multi-positive views (Tian et al., 2023). These methods replace or globally supplement the source and allocate no budget over an observed skewed corpus. DiffAug, the closest iterative predecessor, trains a semantic encoder and a conditional diffusion generator jointly so that generated positives improve contrastive learning (Zang et al., 2024). BRIDGE holds the generator fixed and asks where it should intervene, and our DiffAug control adapts its objective to our source.

Data engines and external priors. Industrial data engines close a loop between a deployed model and its training data. AIDE uses vision-language and language models to diagnose detector failures, curate and auto-label data and verify the result (Liang et al., 2024), and platforms such as Omniverse Replicator supply infrastructure for iterating such pipelines (Worker and Pathak, 2022). Both operate from supervised task failures with annotation in the loop. Our AIDE-style control clusters frozen CLIP features, selects inversely to cluster occupancy, captions the selection and generates from text, and a captioned SDEdit control isolates the value of keeping the image as a condition. A frozen generator also carries knowledge from its own pretraining data. DreamTeacher distills generative features into discriminative backbones (Li et al., 2023), DIFT shows that diffusion activations encode semantic correspondence (Tang et al., 2023), and REPA transfers features in the other direction into a diffusion model (Yu et al., 2025), so distilling the generator’s features directly is a necessary control, which we report. Appendix A tabulates the closest methods by supervision, acquisition signal, data action and feedback.

3 THEORETICAL FRAMEWORK

We model targeted repair as budgeted allocation over coherent regions of representation space. A region is valuable when additional support reduces its contribution to downstream risk, and a repair is useful only when the generator preserves the semantic content of its anchor, so the policy must weigh support deficit against repairability and redundancy. Let $r_k(z; Z)$ be the distance from z to its k th nearest neighbor in Z and let A be a finite set of valid variants generated from z . Proofs are in Appendix C.

Proposition 1 (Monotone local support). *For any finite A , $r_k(z; Z \cup A) \leq r_k(z; Z)$. The inequality is strict when an added point enters the first k neighbors of z and displaces a neighbor at strictly larger distance.*

To connect support to learning, partition the representation into regions j with current effective support n_j and model the regional contribution to downstream error by

$$\mathcal{B}(\mathbf{m}) = \sum_j p_j a_j (n_j + m_j + \kappa)^{-\gamma}, \quad (1)$$

where p_j is the transfer-relevant mass of region j , a_j its difficulty, m_j the number of valid repairs allocated to it, and $\kappa, \gamma > 0$, the usual diminishing-returns form for learning from additional samples. Write $\Delta_j(m_j) = p_j a_j [(n_j + m_j + \kappa)^{-\gamma} - (n_j + m_j + 1 + \kappa)^{-\gamma}]$ for the reduction from one more valid repair.

Proposition 2 (Sparse-region marginal value). *$\Delta_j(m_j)$ is positive and strictly decreasing in $n_j + m_j$. Between two regions of equal transfer mass and difficulty, the region with less effective support has the larger marginal value.*

A finite budget should therefore be spent away from well-supported regions. Sparsity alone is not sufficient, because generation fails on corrupt or incoherent anchors. Let d be a sparsity score, $b(d)$ the marginal value of a valid repair at that score, $\rho(d)$ the probability of a faithful variant, and $\lambda \geq 0$ the cost of an unfaithful one. Expected repair utility is

$$u(d) = \rho(d) b(d) - (1 - \rho(d)) \lambda. \quad (2)$$

Proposition 3 (Interior recoverable-support optimum). *Suppose b and ρ are positive and differentiable on $[d_{\min}, d_{\max}]$ with $b'(d) > 0$ and $\rho'(d) < 0$, and define $g(d) = b'(d)/(b(d) + \lambda) + \rho'(d)/\rho(d)$. If g is strictly decreasing with $g(d_{\min}) > 0$ and $g(d_{\max}) < 0$, then u has a unique maximizer d^* in the interior, characterized by $g(d^*) = 0$.*

The condition compares the relative growth of support value against the relative decay of repairability, and its monotonicity assumption on g is close to assuming the conclusion, so Proposition 6 in the appendix discharges it for $b(d) = b_0 d^\beta$ with $\beta \in (0, 1]$ and $\rho(d) = \rho_0 e^{-\gamma d}$, where g is strictly decreasing for every $\lambda \geq 0$ and $d^* = \beta/\gamma$ at $\lambda = 0$. The optimum moves outward when support value is closer to linear and inward when fidelity decays faster. BRIDGE operationalizes the interior preference with a trimmed score band and estimates neither parameter. The remaining results concern repetition. Consider a separable objective $U(\mathbf{m}) = \sum_j U_j(n_j + m_j)$ with each U_j increasing and discretely concave and a budget $\sum_j m_j = B$.

Proposition 4 (Adaptive allocation under diminishing returns). *For integer allocations, repeatedly assigning one repair to the region with the largest current marginal gain maximizes $U(\mathbf{m})$. A fixed allocation becomes suboptimal once a region's marginal gain falls below that of another region.*

Proposition 5 (Single-stage insensitivity). *Let $\mathbf{m}$ and $\mathbf{m}'$ be any two allocations of the same budget B under Equation 1, computed from one representation and applied without re-estimation. Then*

$$|\mathcal{B}(\mathbf{m}) - \mathcal{B}(\mathbf{m}')| \leq B \gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}, \quad (3)$$

which tends to zero as $B/n_{\min} \rightarrow 0$ for the least-supported region $n_{\min}$.

A policy that re-estimates support after each intervention follows the changing marginal values of Proposition 4, while a frozen policy keeps spending according to an ordering the first intervention has invalidated. Proposition 5 bounds what any rule can achieve when applied once. With 2,500 generated

images against a source of about 10,000, it predicts that no acquisition rule separates from any other after a single stage, and Section 6.4 finds every score band within 1.32 points. The separation over five stages lies outside the single-stage model, because re-estimation makes the support entering each score depend on earlier allocations, and we claim no matching lower bound for the repeated regime. Equation 1 and the monotone behavior of b and ρ are modeling assumptions, and Section 6 tests their observable predictions, an interior optimum, a large gap to uniform and extreme-tail acquisition, and a null effect of the rule at one stage.

4 BRIDGE

4.1 LOOP AND SCORING

Let $D^{(0)}$ be an unlabeled source set and f_θ an SSL encoder. BRIDGE alternates pretraining and repair over C cycles, and at cycle c the encoder is $f_{\theta^{(c)}}$, the source set $D^{(c)}$ and its embedding set $Z^{(c)}$. After each training stage, scoring embeds every current source image and assigns a label-free sparsity score, selection turns that distribution into an anchor set under a generation budget B , and repair conditions the frozen generator on the anchors and appends the variants. The encoder is the only learned component, so the loop needs no labels and applies to any objective that yields embeddings. Every cycle continues from the previous encoder with a fresh optimizer schedule and retains all earlier images, so a cycle costs one pass over the grown source set and never a pretraining run from scratch.

After training on $D^{(c)}$ we embed each image, $z_i^{(c)} = f_{\theta^{(c)}}(x_i)$, normalize to unit length, and compute

$$d_i^{(c)} = \frac{1}{k} \sum_{z \in \mathcal{N}_k(z_i^{(c)}; Z^{(c)} \setminus \{z_i^{(c)}\})} \|z_i^{(c)} - z\|_2^2, \quad (4)$$

where $\mathcal{N}_k(z; Z)$ is the set of k nearest neighbors of z in Z , so a large score marks an image in a sparse part of the current representation. We use $k = 100$, and k between 10 and 200 moves the seven-task average by about one point (Section 6.3).

4.2 SPARSE AND RECOVERABLE ANCHOR SELECTION

The extreme tail is populated by artifacts, degenerate crops and images the generator cannot vary faithfully, and uniform sampling spends most of its budget in regions that are already dense. The selector targets a sparse region that is still internally coherent in three steps (Figure 2). It restricts attention to $U^{(c)} = \{i \mid q_{0.75}^{(c)} \leq d_i^{(c)} \leq q_{0.99}^{(c)}\}$ and builds a histogram over that band whose modal bin has center $m^{(c)}$. Given a candidate-pool multiplier α , it forms the mode-centered pool $C^{(c)} \subseteq U^{(c)}$ of size αB that minimizes $\sum_{i \in C} |d_i^{(c)} - m^{(c)}|$. It then selects the anchor set $S^{(c)}$ of size B by greedy farthest-point sampling in embedding space from that pool. The band excludes the scores where faithful repair is least reliable, the mode favors a region with repeated empirical support over isolated endpoints, and farthest-point sampling keeps the budget off near-duplicates. We use $\alpha = 4$ and $B = 500$. Section 6.3 shows that the lower cutoff can move from the 75th to the 1st percentile without measurable change, while uniform and top-tail acquisition cost 6.6 to 8.0 points over five cycles.

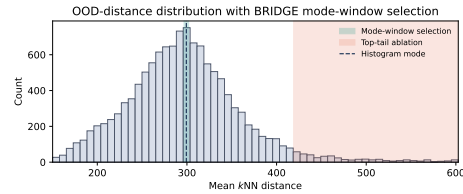


Figure 2: The score distribution of one ImageNet-100-LT run. The mode window centers the candidate pool on the modal bin of the 75th to 99th percentile band (green). The top-tail control spends the same budget in the shaded extreme tail.

4.3 LOCAL GENERATIVE REPAIR AND CONTINUED PRETRAINING

For each anchor we generate $G = 5$ variants with a frozen image-to-image diffusion model g_ϕ , conditioning on the anchor image under an empty text prompt so that the repair signal comes from the image and never from a class name, and set $D^{(c+1)} = D^{(c)} \cup \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$.

The noise ϵ_j indexes independent translations of one anchor, so variants keep coarse structure while changing texture, pose, color or composition, and no generated-image filter is applied. The main configuration is Stable Diffusion 3 Medium with 20 denoising steps, strength 0.6 and output resolution equal to the training crop. With an empty prompt the guidance scale is inert, since at a fixed seed scales 3.0, 5.0 and 7.5 produce bit-identical images and scale 1.0 differs by a mean absolute pixel error of 0.0016 at 49.7 dB. Five training stages contain four repairs, so a full run adds 10,000 images to a source of about 10,000. The score estimates support deficit, the mode window enforces recoverability, farthest-point sampling approximates coverage and cyclic re-estimation reallocates the budget after every intervention, the four requirements of Section 3. Algorithm 1 in the appendix gives the loop.

5 EXPERIMENTAL SETUP

Sources, targets and protocol. We pretrain on five unlabeled source regimes, the long-tailed ImageNet-100-LT (Liu et al., 2019), CIFAR-10-LT and CIFAR-100-LT (Krizhevsky, 2009), and 10k subsets of the web corpora PASS (Asano et al., 2021) and DiffusionDB (Wang et al., 2023). Transfer is measured on seven benchmarks, CIFAR-10-LT, CIFAR-100-LT, Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), Oxford Flowers (Nilsback and Zisserman, 2008), Oxford-IIIT Pets (Parkhi et al., 2012) and ImageNet-100-LT, by linear probe and by k NN classification on frozen features. Every number is a three-seed mean and standard deviation, and paired comparisons share seeds and source checkpoints. ResNet-50 is the default backbone because it is common to the baselines, and BRIDGE trains five cycles of 100 epochs with $k = 100$, $B = 500$ and $G = 5$. DINO and MAE follow their published ViT recipes and never run on a convolutional backbone. Because the downstream tasks are heterogeneous, we report per-task tables and use the seven-task average as a summary, and Appendix D gives implementation details.

Compared methods. The source-regime comparison covers source-matched SimCLR, TS, SDCLR and BRIDGE, with hybrids in the appendix. All ablations use ImageNet-100-LT source pretraining, the same downstream suite and three seeds, and vary one component at a time. The generation ablation includes oracle re-population, which returns the exact real images withheld when the long tail was constructed and so places true support where the method can only estimate that it is missing. That arm is the ceiling available to any repair operator at this budget and is read as an upper bound throughout. The one-stage controls of Section 6.4 and Appendix A branch from one source checkpoint, add the same number of images and receive the same optimizer updates.

6 RESULTS

6.1 TRANSFER ACROSS SOURCE REGIMES

Table 1 reports the three label-derived long-tailed sources, and BRIDGE has the best seven-task average in all three. Under CIFAR-100-LT it raises the average from 23.8 for source-matched SimCLR to 34.4 and is best on every downstream dataset, under ImageNet-100-LT from 39.8 to 42.5 with TS competitive only on the in-domain target, and under CIFAR-10-LT from 29.1 to 34.2. Under k NN evaluation the ranking holds, 31.8 to 33.6, 24.0 to 28.7 and 20.9 to 27.0 (Appendix Table 8), so the gain is present in the neighborhood structure of the features without a learned probe.

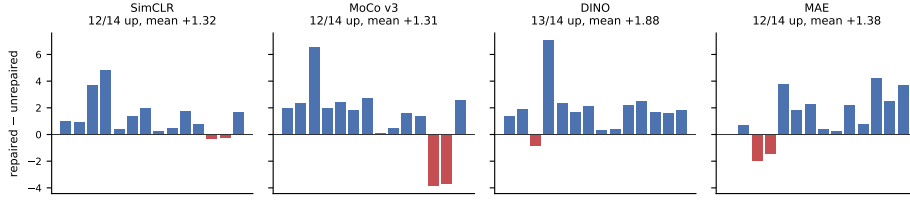
The two web sources behave the same way (Appendix Table 7). On PASS-10k, a natural web corpus, BRIDGE improves the linear-probe average over SimCLR from 36.8 to 39.9 and the k NN average from 29.3 to 30.6, and on DiffusionDB-10k, whose images are themselves synthetic, from 37.7 to 40.1 and from 28.2 to 29.9. Since the DiffusionDB source already carries a generator’s stylistic bias, the improvement there cannot be attributed to harvesting natural web diversity.

6.2 BEYOND THE DEFAULT PROTOCOL

Figure 3 tests whether the effect is specific to SimCLR on ResNet-50 or to frozen-feature evaluation. Four objectives run paired on ViT-S under one schedule and one repair budget, covering contrastive, momentum-contrast, self-distillation and masked-reconstruction pretraining, with both arms of each pair branching from one source checkpoint and receiving the same post-branch updates. Counting linear-probe and k NN accuracy on seven datasets gives 14 measurements per objective. BRIDGE

Table 1: Linear-probe transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.8 $\pm$ 0.2	49.3 $\pm$ 0.2	16.4 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.5 $\pm$ 0.7
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	78.1 $\pm$ 0.3	49.8 $\pm$ 0.4	9.2 $\pm$ 0.8	17.7 $\pm$ 1.0	18.2 $\pm$ 1.1	27.4 $\pm$ 2.7	39.0 $\pm$ 0.6	34.2 $\pm$ 1.0
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	74.2 $\pm$ 0.5	48.1 $\pm$ 0.4	8.3 $\pm$ 1.0	18.2 $\pm$ 0.7	24.4 $\pm$ 2.3	27.6 $\pm$ 0.7	39.8 $\pm$ 1.1	34.4 $\pm$ 1.0

Figure 3: Repair against no repair for four objectives on ViT-S, one bar per downstream measurement (linear probe and k NN on each of seven datasets), from paired runs at three seeds. Blue bars are improvements. The two losses under MoCo v3 are CIFAR-10 and CIFAR-100 under k NN.

improves 12 of 14 for SimCLR, 12 for MoCo v3, 13 for DINO and 12 for MAE, with mean changes of +1.32, +1.31, +1.88 and +1.38 points and a two-sided sign test at $p \leq .013$ in every case. The largest single gains are on Pets, +7.06 for DINO and +4.80 for SimCLR under linear probing (Appendix Tables 25 and 26).

The same source representations improve after end-to-end adaptation. With 1 and 10 percent of the ImageNet-100-LT labels, full fine-tuning gains 1.97 ± 0.78 and 3.17 ± 1.66 points over three paired seeds, and PASCAL VOC semantic segmentation improves from 12.78 ± 0.79 to 15.20 ± 0.86 mIoU, a paired gain of 2.42 ± 0.27 .

6.3 ABLATIONS AND CONTROLS

Selection rule. Table 2 compares three mode windows with different lower cutoffs against top-tail and uniform acquisition over five cycles, all with the same generator, budget and updates. The three windows reach 42.1 to 42.3 and do not differ detectably, while uniform reaches 35.5 ± 1.2 and top-tail 34.3 ± 1.2 , and every window is ahead of both after Holm correction over the eight comparisons at adjusted $p \leq .011$. The exact lower percentile is therefore not the operative choice, and we do not claim that 75 to 99 is optimal. Trimming the extreme tail and diversifying within the remaining sparse band carries the 6.6 to 8.0 points.

Table 2: Selection and cycle ablations on ImageNet-100-LT, seven-task averages over three seeds, one component varied at a time under the same generator, budget and updates.

Group	Setting	Linear probe	k NN
Selection	Mode-window q1–99	42.1 $\pm$ 0.7	33.4 $\pm$ 0.3
	Mode-window q50–99	42.2 $\pm$ 0.9	33.6 $\pm$ 0.3
	Mode-window q75–99	42.3 $\pm$ 0.7	33.6 $\pm$ 0.5
	Uniform	35.5 $\pm$ 1.2	28.4 $\pm$ 0.6
	Top-tail	34.3 $\pm$ 1.2	27.3 $\pm$ 0.6
Schedule	2 cycles	35.0 $\pm$ 1.4	27.9 $\pm$ 0.5
	5 cycles	42.3 $\pm$ 0.7	33.6 $\pm$ 0.5
	10 cycles	34.1 $\pm$ 1.7	28.1 $\pm$ 0.6

Is the gain transfer from the generator? A frozen generator trained on external data carries information the source corpus lacks, so an improvement could reflect distillation from it and nothing about where the budget was spent. The five arms of the selection block call the identical Stable Diffusion 3 checkpoint at identical settings, add the same number of images and receive the same updates, and only the anchors differ. An account in which the generator’s prior supplies the benefit predicts no spread across those arms, and the observed spread is 8.0 points. A direct control in Appendix Table 4 distills the generator’s features into the encoder without any generated image and does not reproduce the effect of targeted repair, while adding BRIDGE to that distillation recovers it.

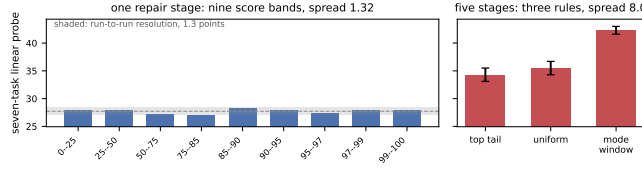


Figure 4: One repair stage against five. Left, nine bands of the score distribution each receive the same budget after a single stage, and all lie inside the shaded envelope, the run-to-run resolution measured from two sensitivity rows that are the same experiment. Right, three acquisition rules under five stages at the same repair volume and optimizer budget.

Generation mechanism. Oracle re-population reaches 40.7 on the seven-task average against 41.5 for generated repair and 35.0 for FLUX at its original sampling settings (Appendix Table 13). Generated repair lands within a point of an operator that inserts the true missing images, so estimating where support is missing costs little relative to knowing it, and the FLUX gap is a sampling-budget artifact. Under matched settings FLUX.1-schnell at guidance 3 reaches 19.68 ± 0.81 on the four-task diagnostic against 19.70 ± 0.35 for SD3 at 20 steps (Appendix Table 5), so no generator is intrinsically stronger in our setting. Strength between 0.3 and 0.9 moves that diagnostic by one point and step count from 6 to 20 by 1.1.

Sensitivity, cycles and backbone. Appendix Table 21 varies the candidate-pool multiplier, upper cutoff, neighborhood size, distance metric and selection strategy one at a time under a single repair stage, and over $k \in \{10, 25, 50, 100, 200\}$ the seven-task average moves between 26.80 and 27.84. The table carries its own resolution scale, since cosine scoring rescales the normalized L_2 score by a positive constant and selects the identical anchor set, and those two rows differ by up to 1.3 points. At fixed total repair volume and optimizer budget, five cycles improve the seven-task averages over two by 7.3 points under linear probing and 5.7 under k NN and remain stronger than ten, so more frequent repair with less learning between repairs does not help monotonically. Across backbones at the default protocol, ResNet-50 leads at 42.3 against 34.8 for ViT-B, while the paired ViT-S experiments above show that the repair effect is present on transformers.

Cost of the loop. Against the no-repair run at an identical update budget, the loop raises wall-clock time by 29 to 46 percent across the three source regimes, almost all of it diffusion sampling (Appendix E). The cost scales with the anchor budget and the variants per anchor, is independent of the cycle count at fixed total budget, and stays fixed while the pretraining cost grows with the source. It is unoptimized. The sampler runs at its default step count, variants are never cached, and a distilled few-step generator would reduce the dominant term directly.

6.4 WHERE THE ACQUISITION RULE ACQUIRES ITS EFFECT

Figure 4 spends one identical budget on each of nine bands of the score distribution, from the densest quartile to the top percentile, and changes nothing else. The nine seven-task averages lie between 26.95 and 28.27, a spread of 1.32 points, the run-to-run resolution of the pipeline. After a single stage the densest quartile and the extreme tail are indistinguishable as places to spend the budget, and the selector sweep and the alternative pipelines behave the same way. Appendix Table 4 places BRIDGE beside a label-free adaptation of TADA, an AIDE-style vision-language pipeline, inverse-cluster-frequency acquisition, a DiffAug adaptation and direct feature distillation under the same single-stage protocol, and the six comparable pipelines span 1.67 points. Under five stages the same three rules separate by 8.0 points, and the two protocols differ only in how often the acquisition signal is recomputed. A single batch of 2,500 well-placed images is worth no more than a badly placed one, and the value of a targeted rule accumulates because every intervention changes the representation that chooses the next, as Proposition 4 describes and Proposition 5 predicts. The interior optimum of Proposition 3 is therefore undetectable in one stage and appears only through the loop.

6.5 REPRESENTATION GEOMETRY

All 500 anchors chosen at cycle 0 have radii in the 75th to 99th percentile band, and the score distribution stays right-skewed with a growing scale over cycles (Appendix Figure 6), so the band is re-derived from the current representation each time (Figure 5). Under the frozen cycle-0 encoder,

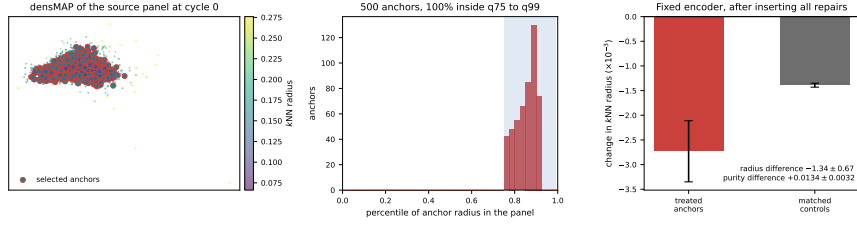


Figure 5: Where the budget lands and what it inserts. Left, densMAP projection of the ImageNet-100-LT source panel under the shared cycle-0 encoder, coloured by k NN radius, with the cycle-0 anchors edged in red. The projection is illustrative, since two dimensions cannot preserve the radii that define the score. Middle, the percentile of every anchor’s radius within the panel. Right, under the frozen cycle-0 encoder, change in k NN radius of treated anchors and of untreated controls matched within class on initial radius after all repairs are inserted, with the paired differences in radius and neighborhood label purity. Three seeds.

treated anchors lose more local radius than untreated controls matched on initial radius and their neighborhoods become purer, which is Proposition 1 measured directly.

On a fixed panel of original images, MoCo v3 on ViT-S gains 21.97 ± 1.58 in effective rank, 0.296 ± 0.019 in spectral entropy and 3.80 ± 0.80 points of 1-NN accuracy after repair, and the first two survive Holm correction. Local support over such a panel becomes more even for DINO, whose Gini coefficient of the k NN radius falls on all three seeds against the unrepaired arm at the same optimizer stage, from 0.1235 ± 0.0121 to 0.1089 ± 0.0093 , a paired change of -0.0147 ± 0.0030 at $p = .014$. Neither global statistic is universal, since SimCLR improves transfer while its effective rank falls, so rank is an objective-dependent diagnostic and the local intervention test the primary evidence.

7 LIMITATIONS

BRIDGE depends on informative encoder neighborhoods in the first cycle and on faithful local generation, and its frozen generator imports the coverage and biases of its own pretraining data. The score band, neighborhood size and budget are operating choices whose defaults transferred across five sources without tuning, and no optimum has been shown for any of them. The fixed-encoder test verifies where support is inserted without establishing how every learned neighborhood evolves, and the learned-space local effect is mixed for SimCLR. Later cycles may select generated images as anchors, no generated-image filter is applied, and a benefit for every objective and target remains unshown. Effectiveness beyond 10k-image web subsets, and whether the fixed budget must grow with the source, remain open.

8 CONCLUSION

Skewed-data SSL can be treated as a feedback problem in which the learner allocates a finite budget to change its own future source distribution. Under diminishing returns to local support and fidelity that decays in the extreme tail, the budget belongs in a sparse interior band and should be reallocated after each repair. BRIDGE implements that policy without labels and improves transfer over source-matched baselines and over uniform, extreme-tail, one-shot and language-mediated alternatives on five sources, four objectives and two backbones. Where the budget is spent has no measurable effect for one intervention and an effect of 8.0 points once the representation that chooses the target has itself been repaired.

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A COMPARISON WITH RELATED ACQUISITION METHODS

Table 3 makes the comparison axes explicit. Classical oversampling and TADA receive labels or supervised learning signals. DOPING uses latent rarity, targets anomaly detection and does not close the loop through a changing SSL representation. GenRep and StableRep make synthetic images or latent neighborhoods the source of contrastive views without allocating a repair budget over an observed skewed corpus. DiffAug is the closest iterative SSL antecedent, and it jointly learns a generator for generic positive views without asking where a frozen generator should intervene. AIDE and industrial data-engine systems establish the model-and-data feedback pattern from supervised task failures with VLM and LLM reasoning, annotation and domain-specific verification. BRIDGE studies the label-free case in which local support in the current SSL representation is both the diagnostic and the acquisition signal. Our VLM control adapts the AIDE idea by clustering frozen CLIP features, selecting inversely to cluster occupancy, captioning the selected images and generating from text. The paired text-to-image and captioned SDEdit controls test whether retaining the selected image as a local condition matters.

Table 3: Closest methods and the dimensions relevant to the acquisition problem. Feedback means that the learner or task model changes the next acquisition decision.

Method	Labels or task loss	Acquisition signal	Data action	Feedback	Primary goal
SMOTE (Chawla et al., 2002)	class labels	class minority	interpolate examples	no	supervised imbalance
DOPING (Lim et al., 2018)	no labels	latent edge density	latent oversampling	no	anomaly detection
Concept pruning (Abbas et al., 2024)	no task labels	cluster complexity	remove examples	no	efficient pretraining
DA-Fusion (Trabucco et al., 2024)	labels or few-shot set	class and exemplar	image-conditioned diffusion	no	supervised recognition
TADA (Nguyen et al., 2026)	supervised loss dynamics	slow-learned examples	faithful diffusion variants	no	supervised recognition
GenRep and StableRep (Jahanian et al., 2022; Tian et al., 2023)	prompts or generator latent	global sampling	synthetic SSL views	no	representation learning
DiffAug (Zang et al., 2024)	no labels	learned positive-view objective	jointly learned diffusion views	yes	contrastive learning
AIDE (Liang et al., 2024)	task labels and failures	VLM and LLM failure analysis	curate, generate, annotate	yes	driving detection
BRIDGE	no labels	local support in current SSL space	budgeted local variants	yes	source-support repair

B REPRESENTATION DIAGNOSTICS

The fixed-encoder intervention test freezes the cycle-0 encoder and queries the same original images before and after inserting all generated samples, so any change reflects the inserted points and never a change of the encoder. First-cycle anchors are matched one-to-one within class to original images with the closest initial k NN radius that are never selected in any cycle. Table 6 reports the direct anchor-against-control comparison. The learned-space comparison retrain the encoder after the intervention and evaluates the same original-image panel. Its local effect is mixed for SimCLR,

Table 4: Empirical comparison with the closest acquisition and repair pipelines on ImageNet-100-LT, all under one matched protocol with a single repair stage, identical anchors budget and identical optimizer budget. Four-task transfer averages Cars, Aircraft, Flowers and Pets under a linear probe; representation 1-NN is measured on the source set. No pipeline separates from any other, BRIDGE included, which is the single-stage regime bounded by Proposition 5. The five-cycle comparisons in Section 6.4 are where acquisition rules do separate.

Method	Seeds	Four-task transfer	Representation 1-NN
Uniform + SD3	3	18.33 ± 0.32	28.97 ± 2.11
Top-tail + SD3	3	19.51 ± 0.22	28.63 ± 1.48
Cluster-frequency + SD3	3	18.66 ± 0.10	28.66 ± 2.42
TADA-SSL + SD3	3	20.00 ± 0.98	28.78 ± 2.22
BRIDGE + SD3	3	18.88 ± 0.58	28.69 ± 1.82
Mode-window + conventional augmentation	3	18.52 ± 0.65	28.91 ± 1.96
DiffAug adaptation	3	11.50 ± 1.87	19.39 ± 2.78
DiffAug adaptation + BRIDGE	3	10.42 ± 2.17	20.28 ± 2.64
SD3 feature distillation	3	18.02 ± 0.47	26.27 ± 2.29
SD3 feature distillation + BRIDGE	3	18.87 ± 0.45	29.45 ± 2.26

Table 5: Generator sensitivity on the four-task diagnostic (Cars, Aircraft, Flowers, ImageNet-100-LT) after one repair stage from a common checkpoint, three paired seeds. The original FLUX setting of six steps and guidance 2.5 is the row that produced the 35.0 seven-task average in Table 13. Guidance under an empty prompt gives bit-identical images at 3.0, 5.0 and 7.5.

Generator	Setting	Four-task transfer
FLUX.1-dev	6 steps, guidance 2.5	18.23 ± 0.98
FLUX.1-dev	20 steps	18.78 ± 1.43
FLUX.1-schnell	20 steps	18.85 ± 1.30
FLUX.1-schnell	guidance 3	19.68 ± 0.81
SD3	6 steps	18.58 ± 1.14
SD3	20 steps (default)	19.70 ± 0.35
SD3	guidance 1.0 to 7.5	18.42 to 18.88
SD3	strength 0.3 to 0.9	18.60 to 19.63

while the global response depends on the objective, as Section 6.5 reports. We therefore treat the fixed-encoder result as evidence about where new support is inserted and the learned-space statistics as objective-dependent diagnostics.

Where the temperature-schedule hybrid helps. Combining BRIDGE with TS lowers the seven-task average in every source regime, and it wins 7 of the 70 individual cells. Six of those seven fall on CIFAR-10 and ImageNet-100-LT, the two long-tailed downstream targets, and the largest are +3.3 and +2.8 points on ImageNet-100-LT and CIFAR-10 under a PASS and a CIFAR-10-LT source. The fine-grained targets lose in almost every case. TS reweights the contrastive temperature toward rare samples, which sharpens the head-to-tail boundary that the long-tailed targets score, while BRIDGE adds support that broadens transfer. We read the hybrid as two interventions serving different targets and report the components separately.

C PROOFS FOR THE SUPPORT-ALLOCATION FRAMEWORK

Proposition 6 (Closed-form interior optimum). *Let $b(d) = b_0 d^\beta$ with $b_0 > 0$ and $\beta \in (0, 1]$, and let $\rho(d) = \rho_0 e^{-\gamma d}$ with $\gamma > 0$. Then g is strictly decreasing on $(0, \infty)$ for every $\lambda \geq 0$, so Proposition 3 applies whenever g changes sign on the interval. If $\lambda = 0$ the maximizer is $d^* = \beta/\gamma$, independent of b_0 and ρ_0 .*

Proof of Proposition 1. Sort the distances from z to points in Z and denote their k th order statistic by $r_k(z; Z)$. Adding A preserves every old candidate distance and may introduce smaller ones. The k th order statistic can therefore only decrease. It decreases strictly when an added distance enters the first k positions and displaces a strictly larger distance. $\square$

Table 6: Local support added in the frozen cycle-0 representation, reported as mean $\pm$ standard deviation over three seeds. Controls are matched one-to-one within class on initial k NN radius and are never selected. Negative radius DiD and positive purity DiD favor treated anchors.

	Δr anchor	Δr control	Radius DiD	Purity DiD
Mean $\pm$ SD	-0.00273 ± 0.00062	-0.00139 ± 0.00004	-0.00134 ± 0.00067	$+0.0134 \pm 0.0032$

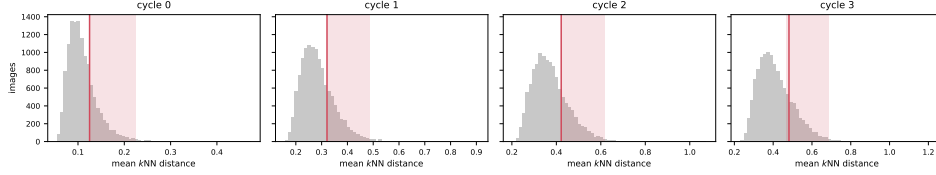


Figure 6: Score distribution over repair cycles for one ImageNet-100-LT run, with the 75th to 99th percentile acquisition band shaded and the modal bin marked. The distribution stays right-skewed while its scale grows, so the band tracks a moving target and the selector re-derives it from the current representation at every cycle.

Proof of Proposition 2. Let $x = n_j + m_j + \kappa$. Since $f(x) = x^{-\gamma}$ is positive, strictly decreasing and strictly convex for $x > 0$, the forward difference $f(x) - f(x+1)$ is positive and strictly decreases with x . Multiplication by $p_j a_j > 0$ preserves both properties. Equal $p_j a_j$ therefore leaves effective support as the sole determinant of the marginal ordering. $\square$

Proof of Proposition 3. Rewrite the utility as $u(d) = \rho(d)(b(d) + \lambda) - \lambda$. Its derivative is

$$u'(d) = \rho(d)(b(d) + \lambda) \left[\frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)} \right] = \rho(d)(b(d) + \lambda) g(d).$$

The factor preceding $g(d)$ is positive. Since g is strictly decreasing and changes sign across the interval, it has exactly one root d^* by continuity. Hence u increases before d^* and decreases after it, making d^* the unique global maximizer. $\square$

Proof of Proposition 6. With $b(d) = b_0 d^\beta$ and $\rho(d) = \rho_0 e^{-\gamma d}$ we have $\rho'(d)/\rho(d) = -\gamma$, so

$$g(d) = \frac{b_0 \beta d^{\beta-1}}{b_0 d^\beta + \lambda} - \gamma = \frac{\beta d^{\beta-1}}{d^\beta + c} - \gamma, \quad c = \lambda/b_0 \geq 0.$$

Write $h(d) = \beta d^{\beta-1}(d^\beta + c)^{-1}$. Differentiating,

$$h'(d) = \frac{\beta d^{\beta-2}[(\beta-1)(d^\beta + c) - \beta d^\beta]}{(d^\beta + c)^2} = \frac{\beta d^{\beta-2}[(\beta-1)c - d^\beta]}{(d^\beta + c)^2}.$$

For $\beta \in (0, 1]$ we have $(\beta-1)c \leq 0 < d^\beta$ for all $d > 0$, so the bracket is strictly negative and $h'(d) < 0$. Hence $g = h - \gamma$ is strictly decreasing for every $\lambda \geq 0$. When $\lambda = 0$ we have $c = 0$ and $g(d) = \beta/d - \gamma$, whose unique root is $d^* = \beta/\gamma$. Since g is positive below it and negative above it, d^* is the maximizer by the argument of Proposition 3. $\square$

Proof of Proposition 4. For region j define the marginal sequence $\delta_{j,t} = U_j(n_j + t) - U_j(n_j + t - 1)$ for $t \geq 1$. Discrete concavity makes each sequence nonincreasing. Any feasible allocation of B repairs selects a prefix of length m_j from each sequence. Greedy allocation repeatedly chooses the largest available next element, and after B steps it has selected the B largest prefix-feasible marginals. If another allocation had larger utility, it would contain a selected marginal smaller than an available unselected marginal, and exchanging them would improve that allocation, which contradicts its optimality. A frozen allocation can fail this rule because it preserves the initial ordering after selected regions have moved to smaller marginals. $\square$

Proof of Proposition 5. Write $f(x) = x^{-\gamma}$. For $x > 0$ the mean value theorem gives $f(x) - f(x+1) = \gamma \xi^{-\gamma-1}$ for some $\xi \in (x, x+1)$, so

$$0 < \Delta_j(m_j) = p_j a_j [f(n_j + m_j + \kappa) - f(n_j + m_j + 1 + \kappa)] \leq p_j a_j \gamma (n_j + \kappa)^{-\gamma-1},$$

using $\xi > n_j + m_j + \kappa \geq n_j + \kappa$ and $m_j \geq 0$. Both $\mathcal{B}(\mathbf{m})$ and $\mathcal{B}(\mathbf{m}')$ are obtained from the common baseline $\mathcal{B}(\mathbf{0})$ by subtracting sums of at most B such marginals, since $\sum_j m_j = \sum_j m'_j = B$. Each sum lies in $[0, B\gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}]$, and the difference of two numbers drawn from an interval is bounded by its length. Monotonicity of $(n + \kappa)^{-\gamma-1}$ in n gives the $n_{\min}$ form. $\square$

D IMPLEMENTATION AND REPRODUCIBILITY

Training schedule. The encoder is warm-started from the previous stage at every cycle and is never retrained from scratch. Each stage uses a fresh optimizer and learning-rate schedule. All earlier real and generated images remain in the source set, so the setting differs from sequential-task learning in which previous data disappear. Five training stages contain four repairs, since generation runs only when another stage follows. The cycle ablation holds the total optimizer update budget and the total number of generated images fixed and compares fewer repairs with more learning between them against more frequent repairs with less.

Core configuration. The main experiments use ResNet-50 encoders, five cycles of 100 epochs, mean k NN distance with $k = 100$, 500 anchors per cycle and five variants per anchor. The optimizer is AdamW with weight decay 10^{-4} , and method-specific learning rates and temperatures follow the settings of each baseline. The DINO arm uses the reference recipe with a ViT-S backbone, two global and eight local crops at 96×96 and a bottleneck projection head. MAE is ViT-only and runs five cycles of 80 epochs.

Scoring and selection. The submitted experiments used raw squared ℓ_2 distance for scoring and normalized features for farthest-point sampling. Because raw feature norms can confound cross-objective comparisons, the controlled experiments use normalized ℓ_2 throughout and report raw ℓ_2 , normalized ℓ_2 , cosine and multiscale- k variants as a robustness study. The mode-window selector restricts the score distribution to the 75th to 99th percentile range, builds a histogram with the automatic bin heuristic, forms a mode-centered candidate pool of size $4B$ and applies greedy farthest-point sampling to select the final B anchors.

Generators. The main method uses Stable Diffusion 3 Medium in image-to-image mode. The generation ablation replaces it with FLUX or with oracle re-population, which adds the withheld real images and runs no generator. In the PASS and DiffusionDB experiments the SD3 batch size is 12 to fit the 24-hour wall-time limit.

A corrected DINO recipe. The DINO arm was rebuilt after we found that an earlier implementation omitted the bottleneck projection head and applied solarization to the wrong crops. Its baseline is competitive with SimCLR and MoCo v3 on the same schedule.

A checkpointing fault and the reruns it forced. Each cycle constructs a fresh trainer whose step counter restarts at zero and stops at the same step limit. The checkpoint callback, reused across cycles, treated a save at a step it had already written as a duplicate and skipped it, so runs configured with a per-cycle step cap published their first-cycle encoder however far later cycles diverged. We found this when two policy conditions reproduced each other to the last bit while their anchor manifests overlapped on 14 of 500 anchors from the second cycle onward. Every paired ViT-S cell behind Figure 3 was rerun with the callback state reset between cycles, 19 runs and 416 wall-clock hours, and we verify before publication that no two conditions in a table share an encoder fingerprint. Runs whose cycles end on an epoch boundary were never affected.

Source data. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. The long-tailed ImageNet-100, CIFAR-10 and CIFAR-100 sources use their standard long-tailed configurations without the 10k restriction.

Seeds and reporting. All claims rest on three-seed averages reported as mean and standard deviation. Paired comparisons share seeds and, where stated, source checkpoints. Multiple comparisons within one table are corrected by the Holm procedure.

Table 7: Linear-probe transfer for the two web-source regimes. PASS is a natural web-image corpus, while DiffusionDB is fully synthetic. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	73.9 $\pm$ 0.3	47.1 $\pm$ 0.4	13.6 $\pm$ 0.7	20.5 $\pm$ 0.7	37.7 $\pm$ 4.0	37.1 $\pm$ 0.7	49.1 $\pm$ 0.5	39.9 $\pm$ 1.0
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	75.9 $\pm$ 0.4	49.9 $\pm$ 0.5	12.8 $\pm$ 1.0	20.0 $\pm$ 0.4	35.7 $\pm$ 1.1	37.7 $\pm$ 0.8	49.1 $\pm$ 0.5	40.1 $\pm$ 0.7

Code availability. An anonymous code release is available at <https://anonymous.4open.science/status/FOMO-0488>.

E COMPUTE RESOURCES AND RUNTIME

Most experiments use one node with two NVIDIA A100 80GB GPUs under a 24-hour wall-time limit. The DINO arm fits on a single 48GB card at the batch size used. The 454 completed runs retained for this paper total approximately 5,471 wall-clock hours. Runs on two-GPU nodes consume roughly twice that in GPU hours, while the paired ViT-S reruns hold a single card each. The 19 reruns that replaced encoders affected by the checkpointing fault account for 416 hours, and the baselines and controls that appear only in the appendix for a further 2,987. These totals exclude failed runs, timeouts, cache migration and pilot jobs, so the project consumed more compute than the successful totals alone.

Representative mean wall-clock times over three seeds are as follows. On ImageNet-100-LT, SimCLR takes 16.9 hours and BRIDGE 21.8. On PASS, 10.2 and 14.9. On DiffusionDB, 11.6 and 16.4. The loop therefore raises wall-clock time by 29 to 46 percent over the corresponding no-repair run, of which the embedding pass is a small fraction and diffusion sampling the remainder. Four repairs add 10,000 images in total, which is 100 percent of a 10k source, 10 percent of a 100k source and 1 percent of a one-million-image source. This describes the relative cost under a fixed budget and does not establish that the fixed budget remains effective at those scales.

F ASSETS AND TERMS OF USE

ImageNet-100-LT is derived from ImageNet, whose access terms restrict the images to non-commercial research and educational use. PASS is released under Creative Commons Attribution 4.0 and DiffusionDB under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers and Oxford-IIIT Pets are used from their official academic releases under their original terms. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation under the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third-party datasets or generated image corpora.

G BROADER IMPACT

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful where balanced data are difficult to collect. The method also relies on large image generators and web-sourced datasets, which carry risks related to license compliance, privacy and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and release no new generators or scraped collections. Any deployment would require a task-specific review of fairness, privacy and content safety.

Table 8: k NN transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream k NN tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	7.0 $\pm$ 0.2	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.4
	BRIDGE+TS	65.4 $\pm$ 0.2	33.6 $\pm$ 0.3	3.1 $\pm$ 0.1	7.1 $\pm$ 0.3	18.2 $\pm$ 0.2	15.6 $\pm$ 0.7	40.8 $\pm$ 0.5	26.3 $\pm$ 0.3
	BRIDGE+SDCLR	36.3 $\pm$ 0.8	13.0 $\pm$ 0.5	1.1 $\pm$ 0.3	2.4 $\pm$ 0.2	9.7 $\pm$ 0.3	5.4 $\pm$ 0.4	13.9 $\pm$ 1.0	11.7 $\pm$ 0.5
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	74.8 $\pm$ 0.4	41.0 $\pm$ 0.4	4.8 $\pm$ 0.5	10.0 $\pm$ 0.2	24.2 $\pm$ 0.6	16.2 $\pm$ 0.7	29.7 $\pm$ 0.3	28.7 $\pm$ 0.4
	BRIDGE+TS	77.6 $\pm$ 0.7	35.3 $\pm$ 0.5	3.1 $\pm$ 0.3	6.6 $\pm$ 0.4	16.3 $\pm$ 0.6	12.5 $\pm$ 1.0	26.7 $\pm$ 0.1	25.5 $\pm$ 0.5
	BRIDGE+SDCLR	34.4 $\pm$ 1.2	11.4 $\pm$ 0.6	1.2 $\pm$ 0.2	2.2 $\pm$ 0.2	8.8 $\pm$ 1.5	5.0 $\pm$ 0.2	11.7 $\pm$ 1.1	10.7 $\pm$ 0.7
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	69.3 $\pm$ 0.5	39.5 $\pm$ 0.1	4.8 $\pm$ 0.6	8.9 $\pm$ 0.1	21.9 $\pm$ 0.1	15.5 $\pm$ 0.1	29.1 $\pm$ 0.7	27.0 $\pm$ 0.3
	BRIDGE+TS	64.3 $\pm$ 0.4	36.6 $\pm$ 0.4	2.9 $\pm$ 0.3	6.3 $\pm$ 0.1	16.4 $\pm$ 0.5	12.3 $\pm$ 0.0	28.2 $\pm$ 0.6	23.9 $\pm$ 0.3
	BRIDGE+SDCLR	31.9 $\pm$ 0.4	10.4 $\pm$ 0.5	0.8 $\pm$ 0.1	2.1 $\pm$ 0.5	7.5 $\pm$ 1.6	4.5 $\pm$ 0.1	9.1 $\pm$ 0.7	9.5 $\pm$ 0.6

H FULL RESULTS

Table 9: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 1.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.8 $\pm$ 0.2	49.3 $\pm$ 0.2	16.4 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.5 $\pm$ 0.7
	BRIDGE+TS (ours)	74.2 $\pm$ 0.5	45.6 $\pm$ 0.8	9.4 $\pm$ 0.5	15.0 $\pm$ 0.2	25.5 $\pm$ 2.5	33.1 $\pm$ 2.8	48.3 $\pm$ 0.4	35.9 $\pm$ 1.1
	BRIDGE+SDCLR (ours)	43.8 $\pm$ 1.7	18.4 $\pm$ 0.7	2.0 $\pm$ 0.3	2.5 $\pm$ 0.8	10.0 $\pm$ 0.8	12.2 $\pm$ 1.3	21.8 $\pm$ 0.5	15.8 $\pm$ 1.1
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	78.1 $\pm$ 0.3	49.8 $\pm$ 0.4	9.2 $\pm$ 0.8	17.7 $\pm$ 1.0	18.2 $\pm$ 1.1	27.4 $\pm$ 2.7	39.0 $\pm$ 0.6	34.2 $\pm$ 1.0
	BRIDGE+TS (ours)	78.6 $\pm$ 0.3	46.1 $\pm$ 0.3	7.5 $\pm$ 0.9	11.1 $\pm$ 0.7	14.5 $\pm$ 0.6	25.1 $\pm$ 2.7	39.6 $\pm$ 0.8	31.8 $\pm$ 0.9
	BRIDGE+SDCLR (ours)	40.9 $\pm$ 0.7	17.9 $\pm$ 0.5	1.4 $\pm$ 0.1	2.2 $\pm$ 0.6	11.0 $\pm$ 2.4	10.0 $\pm$ 2.7	21.0 $\pm$ 0.6	14.9 $\pm$ 1.1
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	74.2 $\pm$ 0.5	48.1 $\pm$ 0.4	8.3 $\pm$ 1.0	18.2 $\pm$ 0.7	24.4 $\pm$ 2.3	27.6 $\pm$ 0.7	39.8 $\pm$ 1.1	34.4 $\pm$ 1.0
	BRIDGE+TS (ours)	73.7 $\pm$ 0.1	47.3 $\pm$ 0.5	7.9 $\pm$ 0.6	14.1 $\pm$ 0.5	18.2 $\pm$ 3.2	28.4 $\pm$ 0.4	41.3 $\pm$ 0.3	33.0 $\pm$ 0.8
	BRIDGE+SDCLR (ours)	42.0 $\pm$ 1.5	16.9 $\pm$ 0.9	1.3 $\pm$ 0.1	2.2 $\pm$ 0.6	10.6 $\pm$ 0.8	8.3 $\pm$ 0.8	20.2 $\pm$ 0.9	14.5 $\pm$ 0.8
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	73.9 $\pm$ 0.3	47.1 $\pm$ 0.4	13.6 $\pm$ 0.7	20.5 $\pm$ 0.7	37.7 $\pm$ 4.0	37.1 $\pm$ 0.7	49.1 $\pm$ 0.5	39.9 $\pm$ 1.0
	BRIDGE+TS (ours)	68.9 $\pm$ 0.6	40.6 $\pm$ 0.7	12.1 $\pm$ 0.4	14.1 $\pm$ 0.5	34.6 $\pm$ 2.9	31.6 $\pm$ 1.5	52.4 $\pm$ 0.6	36.4 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	44.0 $\pm$ 0.9	18.8 $\pm$ 0.9	1.4 $\pm$ 0.3	1.8 $\pm$ 0.9	12.6 $\pm$ 2.6	12.4 $\pm$ 1.3	21.2 $\pm$ 1.1	16.0 $\pm$ 1.1
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	75.9 $\pm$ 0.4	49.9 $\pm$ 0.5	12.8 $\pm$ 1.0	20.0 $\pm$ 0.4	35.7 $\pm$ 1.1	37.7 $\pm$ 0.8	49.1 $\pm$ 0.5	40.1 $\pm$ 0.7
	BRIDGE+TS (ours)	67.8 $\pm$ 0.6	39.1 $\pm$ 0.5	9.4 $\pm$ 1.2	12.4 $\pm$ 0.4	30.3 $\pm$ 3.0	33.7 $\pm$ 1.9	50.8 $\pm$ 0.6	34.8 $\pm$ 1.2
	BRIDGE+SDCLR (ours)	43.8 $\pm$ 0.9	18.5 $\pm$ 0.3	1.3 $\pm$ 0.4	2.5 $\pm$ 0.0	14.3 $\pm$ 1.6	11.4 $\pm$ 0.6	21.9 $\pm$ 0.6	16.2 $\pm$ 0.6

Table 10: Full k NN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	7.0 $\pm$ 0.2	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.4
	BRIDGE+TS (ours)	65.4 $\pm$ 0.2	33.6 $\pm$ 0.3	3.1 $\pm$ 0.1	7.1 $\pm$ 0.3	18.2 $\pm$ 0.2	15.6 $\pm$ 0.7	40.8 $\pm$ 0.5	26.3 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	36.3 $\pm$ 0.8	13.0 $\pm$ 0.5	1.1 $\pm$ 0.3	2.4 $\pm$ 0.2	9.7 $\pm$ 0.3	5.4 $\pm$ 0.4	13.9 $\pm$ 1.0	11.7 $\pm$ 0.5
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	74.8 $\pm$ 0.4	41.0 $\pm$ 0.4	4.8 $\pm$ 0.5	10.0 $\pm$ 0.2	24.2 $\pm$ 0.6	16.2 $\pm$ 0.7	29.7 $\pm$ 0.3	28.7 $\pm$ 0.4
	BRIDGE+TS (ours)	77.6 $\pm$ 0.7	35.3 $\pm$ 0.5	3.1 $\pm$ 0.3	6.6 $\pm$ 0.4	16.3 $\pm$ 0.6	12.5 $\pm$ 1.0	26.7 $\pm$ 0.1	25.5 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	34.4 $\pm$ 1.2	11.4 $\pm$ 0.6	1.2 $\pm$ 0.2	2.2 $\pm$ 0.2	8.8 $\pm$ 1.5	5.0 $\pm$ 0.2	11.7 $\pm$ 1.1	10.7 $\pm$ 0.7
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	69.3 $\pm$ 0.5	39.5 $\pm$ 0.1	4.8 $\pm$ 0.6	8.9 $\pm$ 0.1	21.9 $\pm$ 0.1	15.5 $\pm$ 0.1	29.1 $\pm$ 0.7	27.0 $\pm$ 0.3
	BRIDGE+TS (ours)	64.3 $\pm$ 0.4	36.6 $\pm$ 0.4	2.9 $\pm$ 0.3	6.3 $\pm$ 0.1	16.4 $\pm$ 0.5	12.3 $\pm$ 0.0	28.2 $\pm$ 0.6	23.9 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	31.9 $\pm$ 0.4	10.4 $\pm$ 0.5	0.8 $\pm$ 0.1	2.1 $\pm$ 0.5	7.5 $\pm$ 1.6	4.5 $\pm$ 0.1	9.1 $\pm$ 0.7	9.5 $\pm$ 0.6
PASS-10k	SimCLR	67.2 $\pm$ 0.2	37.0 $\pm$ 0.3	5.4 $\pm$ 0.3	10.3 $\pm$ 0.5	30.4 $\pm$ 0.0	16.7 $\pm$ 0.5	38.3 $\pm$ 0.3	29.3 $\pm$ 0.3
	TS	65.2 $\pm$ 0.7	34.6 $\pm$ 0.3	4.4 $\pm$ 0.2	7.9 $\pm$ 0.2	24.5 $\pm$ 0.1	14.4 $\pm$ 0.4	39.3 $\pm$ 0.4	27.2 $\pm$ 0.3
	SDCLR	32.4 $\pm$ 1.5	11.5 $\pm$ 0.8	1.3 $\pm$ 0.2	2.0 $\pm$ 0.1	8.6 $\pm$ 0.7	5.2 $\pm$ 0.2	11.1 $\pm$ 0.8	10.3 $\pm$ 0.6
	BRIDGE (ours)	67.5 $\pm$ 0.3	36.9 $\pm$ 0.3	5.8 $\pm$ 0.5	10.7 $\pm$ 0.4	33.7 $\pm$ 0.2	17.8 $\pm$ 0.6	41.4 $\pm$ 0.8	30.6 $\pm$ 0.4
	BRIDGE+TS (ours)	58.8 $\pm$ 0.5	27.9 $\pm$ 0.5	3.0 $\pm$ 0.2	6.6 $\pm$ 0.1	23.0 $\pm$ 0.3	13.4 $\pm$ 0.4	38.8 $\pm$ 0.8	24.5 $\pm$ 0.4
	BRIDGE+SDCLR (ours)	37.1 $\pm$ 2.0	13.6 $\pm$ 1.0	1.6 $\pm$ 0.2	2.4 $\pm$ 0.2	10.1 $\pm$ 0.5	6.3 $\pm$ 0.3	13.1 $\pm$ 0.6	12.0 $\pm$ 0.7
DiffusionDB-10k	SimCLR	67.3 $\pm$ 0.1	36.2 $\pm$ 0.4	4.3 $\pm$ 0.2	8.8 $\pm$ 0.1	26.8 $\pm$ 0.4	17.3 $\pm$ 0.1	36.7 $\pm$ 0.8	28.2 $\pm$ 0.3
	TS	64.7 $\pm$ 0.4	33.6 $\pm$ 0.3	3.1 $\pm$ 0.5	7.3 $\pm$ 0.3	23.2 $\pm$ 0.4	17.0 $\pm$ 0.0	37.8 $\pm$ 0.6	26.7 $\pm$ 0.4
	SDCLR	32.6 $\pm$ 1.2	10.0 $\pm$ 0.3	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.1 $\pm$ 0.7	4.8 $\pm$ 0.4	10.6 $\pm$ 0.5	9.7 $\pm$ 0.5
	BRIDGE (ours)	68.3 $\pm$ 0.2	37.4 $\pm$ 0.4	5.4 $\pm$ 0.3	9.8 $\pm$ 0.1	30.4 $\pm$ 0.2	18.6 $\pm$ 0.2	39.4 $\pm$ 1.0	29.9 $\pm$ 0.3
	BRIDGE+TS (ours)	57.5 $\pm$ 0.8	26.6 $\pm$ 0.3	3.0 $\pm$ 0.4	6.5 $\pm$ 0.1	21.7 $\pm$ 0.8	15.5 $\pm$ 0.4	37.0 $\pm$ 0.4	24.0 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	35.2 $\pm$ 0.7	11.1 $\pm$ 0.3	1.0 $\pm$ 0.1	2.2 $\pm$ 0.1	8.1 $\pm$ 1.0	5.0 $\pm$ 0.3	11.3 $\pm$ 0.7	10.6 $\pm$ 0.5

Table 11: Pretraining objective ablation on the default ResNet-50 backbone, full linear-probe results. DINO is excluded here because its reference recipe is defined for ViT backbones; the DINO comparison is reported with ViT-S in Table 25.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
MoCo	50.5 $\pm$ 1.0	20.6 $\pm$ 2.7	3.0 $\pm$ 0.2	5.7 $\pm$ 0.5	11.5 $\pm$ 3.0	15.3 $\pm$ 0.3	28.2 $\pm$ 0.7	19.3 $\pm$ 1.2

Table 12: Pretraining objective ablation on the default ResNet-50 backbone, full k NN results. DINO is excluded here because its reference recipe is defined for ViT backbones; the DINO comparison is reported with ViT-S in Table 26.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
MoCo	58.0 $\pm$ 0.4	26.7 $\pm$ 0.5	3.0 $\pm$ 0.2	5.4 $\pm$ 0.2	16.3 $\pm$ 0.9	12.2 $\pm$ 0.6	25.1 $\pm$ 0.5	21.0 $\pm$ 0.5

Table 13: Augmentation mechanism ablation with full linear-probe results. Oracle re-population is not a competing generator: it re-adds the exact real images withheld when the long tail was built, so it has access to the identity and location of the missing support, which the method only estimates. It is the upper bound on what any repair operator can achieve at this budget, and a generator that approaches it is recovering most of the benefit of the withheld real data without access to it.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	75.0 $\pm$ 0.3	48.5 $\pm$ 0.2	16.3 $\pm$ 0.8	20.7 $\pm$ 0.4	35.7 $\pm$ 0.9	41.8 $\pm$ 0.8	52.4 $\pm$ 0.7	41.5 $\pm$ 0.6
FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	10.3 $\pm$ 1.3	17.5 $\pm$ 0.9	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
Oracle re-population (upper bound)	74.8 $\pm$ 0.4	47.8 $\pm$ 0.7	13.1 $\pm$ 0.1	20.5 $\pm$ 0.5	40.3 $\pm$ 6.1	38.1 $\pm$ 1.5	50.3 $\pm$ 0.4	40.7 $\pm$ 1.4

Table 14: Augmentation mechanism ablation with full k NN results. Oracle re-population is not a competing generator: it re-adds the exact real images withheld when the long tail was built, so it has access to the identity and location of the missing support, which the method only estimates. It is the upper bound on what any repair operator can achieve at this budget, and a generator that approaches it is recovering most of the benefit of the withheld real data without access to it.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	68.6 $\pm$ 0.1	37.2 $\pm$ 0.2	6.4 $\pm$ 0.6	11.5 $\pm$ 0.2	35.2 $\pm$ 0.3	24.3 $\pm$ 0.6	46.8 $\pm$ 0.7	32.9 $\pm$ 0.4
FLUX	65.8 $\pm$ 1.0	35.4 $\pm$ 0.8	5.7 $\pm$ 0.7	9.4 $\pm$ 0.9	27.7 $\pm$ 1.0	18.8 $\pm$ 1.3	38.7 $\pm$ 1.2	28.8 $\pm$ 1.0
Oracle re-population (upper bound)	68.8 $\pm$ 0.2	37.7 $\pm$ 0.3	6.6 $\pm$ 0.5	11.4 $\pm$ 0.5	32.7 $\pm$ 0.1	23.6 $\pm$ 0.6	44.4 $\pm$ 0.4	32.2 $\pm$ 0.4

Table 15: Selection rule ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	75.6 $\pm$ 0.1	49.2 $\pm$ 0.2	16.9 $\pm$ 0.7	21.9 $\pm$ 0.4	34.6 $\pm$ 2.9	43.5 $\pm$ 0.7	52.8 $\pm$ 0.1	42.1 $\pm$ 0.7
Mode-window q50-99 + diversity	75.5 $\pm$ 0.1	49.3 $\pm$ 0.4	15.2 $\pm$ 0.3	22.4 $\pm$ 0.6	35.5 $\pm$ 1.1	43.8 $\pm$ 3.1	53.4 $\pm$ 0.9	42.2 $\pm$ 0.9
Mode-window q75-99 + diversity (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
Top-tail	70.8 $\pm$ 0.5	43.1 $\pm$ 1.0	10.0 $\pm$ 1.7	15.9 $\pm$ 1.3	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4	34.3 $\pm$ 1.2
Uniform	71.5 $\pm$ 0.2	43.8 $\pm$ 0.7	9.3 $\pm$ 0.9	17.5 $\pm$ 1.2	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9	35.5 $\pm$ 1.2

Table 16: Selection rule ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	69.2 $\pm$ 0.3	37.7 $\pm$ 0.2	6.6 $\pm$ 0.2	11.4 $\pm$ 0.2	35.9 $\pm$ 0.2	25.0 $\pm$ 0.5	48.3 $\pm$ 0.3	33.4 $\pm$ 0.3
Mode-window q50-99 + diversity	68.7 $\pm$ 0.1	38.0 $\pm$ 0.2	7.3 $\pm$ 0.2	11.3 $\pm$ 0.2	36.5 $\pm$ 0.1	24.8 $\pm$ 0.8	48.5 $\pm$ 0.4	33.6 $\pm$ 0.3
Mode-window q75-99 + diversity (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
Top-tail	65.7 $\pm$ 0.8	34.4 $\pm$ 0.2	4.1 $\pm$ 0.4	9.6 $\pm$ 0.2	23.9 $\pm$ 0.9	18.1 $\pm$ 0.8	35.4 $\pm$ 1.2	27.3 $\pm$ 0.6
Uniform	66.1 $\pm$ 1.0	35.7 $\pm$ 0.3	4.9 $\pm$ 0.1	10.2 $\pm$ 0.7	25.3 $\pm$ 0.4	19.0 $\pm$ 0.8	37.4 $\pm$ 0.6	28.4 $\pm$ 0.6

Table 17: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	68.9 $\pm$ 0.6	41.6 $\pm$ 1.3	10.6 $\pm$ 1.2	16.4 $\pm$ 0.4	30.7 $\pm$ 4.5	33.2 $\pm$ 0.9	43.3 $\pm$ 0.6	35.0 $\pm$ 1.4
5 cycles (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7

Table 18: Cycle count ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.3 $\pm$ 0.4	34.2 $\pm$ 0.5	5.0 $\pm$ 0.7	10.0 $\pm$ 0.2	26.1 $\pm$ 0.8	18.7 $\pm$ 0.6	35.9 $\pm$ 0.3	27.9 $\pm$ 0.5
5 cycles (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	10.4 $\pm$ 0.6	25.2 $\pm$ 0.8	18.6 $\pm$ 0.9	36.0 $\pm$ 0.7	28.1 $\pm$ 0.6

Table 19: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	68.8 $\pm$ 0.1	41.7 $\pm$ 0.2	13.5 $\pm$ 0.3	19.7 $\pm$ 0.6	38.7 $\pm$ 3.4	38.4 $\pm$ 2.6	50.2 $\pm$ 0.2	38.7 $\pm$ 1.1
ResNet-50 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	8.8 $\pm$ 0.5	12.2 $\pm$ 0.2	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	8.3 $\pm$ 0.5	14.3 $\pm$ 0.8	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

Table 20: Backbone architecture ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	66.3 $\pm$ 0.2	35.7 $\pm$ 0.3	5.9 $\pm$ 0.2	11.1 $\pm$ 0.4	33.5 $\pm$ 0.1	23.3 $\pm$ 0.5	44.5 $\pm$ 0.7	31.5 $\pm$ 0.4
ResNet-50 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
ViT-S	67.3 $\pm$ 0.4	37.5 $\pm$ 0.3	5.3 $\pm$ 0.2	7.6 $\pm$ 0.1	36.2 $\pm$ 0.7	20.1 $\pm$ 1.0	42.1 $\pm$ 1.0	30.9 $\pm$ 0.5
ViT-B	69.3 $\pm$ 0.8	40.1 $\pm$ 0.8	6.0 $\pm$ 0.2	8.3 $\pm$ 0.2	39.8 $\pm$ 0.8	21.9 $\pm$ 1.4	45.1 $\pm$ 0.7	32.9 $\pm$ 0.7

Table 21: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean $\pm$ standard deviation over three seeds. Cosine scoring rescales the normalized L_2 score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized L_2 row measures the end-to-end run-to-run spread of the pipeline. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Candidate pool $\alpha = 1$	6.46 $\pm$ 0.57	11.37 $\pm$ 1.59	23.53 $\pm$ 3.00	21.92 $\pm$ 2.00	61.94 $\pm$ 0.94	32.73 $\pm$ 0.21	33.92 $\pm$ 1.68
Candidate pool $\alpha = 2$	6.77 $\pm$ 0.63	11.59 $\pm$ 0.40	21.79 $\pm$ 3.22	23.19 $\pm$ 3.93	62.90 $\pm$ 1.06	34.48 $\pm$ 0.11	35.69 $\pm$ 0.87
Candidate pool $\alpha = 4$ (default)	6.31 $\pm$ 0.10	11.17 $\pm$ 3.78	23.53 $\pm$ 3.27	24.18 $\pm$ 2.16	62.11 $\pm$ 0.19	33.39 $\pm$ 0.64	34.18 $\pm$ 0.47
Candidate pool $\alpha = 8$	6.41 $\pm$ 0.28	11.16 $\pm$ 0.78	23.09 $\pm$ 4.35	24.82 $\pm$ 3.29	62.60 $\pm$ 1.79	33.15 $\pm$ 1.01	35.40 $\pm$ 0.87
Upper cutoff $q = 0.95$	6.79 $\pm$ 0.65	11.44 $\pm$ 0.85	19.83 $\pm$ 2.47	25.27 $\pm$ 2.32	62.57 $\pm$ 0.87	33.83 $\pm$ 0.53	35.13 $\pm$ 1.03
Upper cutoff $q = 0.99$ (default)	6.59 $\pm$ 0.17	11.76 $\pm$ 1.54	20.26 $\pm$ 1.73	24.00 $\pm$ 2.20	63.11 $\pm$ 0.75	33.43 $\pm$ 1.02	35.85 $\pm$ 1.15
Upper cutoff $q = 0.995$	6.78 $\pm$ 0.66	10.40 $\pm$ 2.49	22.88 $\pm$ 4.58	22.55 $\pm$ 3.09	62.76 $\pm$ 0.50	33.87 $\pm$ 0.38	35.32 $\pm$ 0.64
Upper cutoff $q = 1.0$	6.31 $\pm$ 0.45	10.94 $\pm$ 1.34	22.44 $\pm$ 1.89	23.37 $\pm$ 4.00	62.85 $\pm$ 1.04	33.91 $\pm$ 0.87	35.51 $\pm$ 1.59
Neighbourhood $k = 10$	6.26 $\pm$ 0.29	10.44 $\pm$ 1.43	22.22 $\pm$ 0.65	24.00 $\pm$ 1.23	62.66 $\pm$ 0.60	33.44 $\pm$ 0.58	35.14 $\pm$ 1.21
Neighbourhood $k = 25$	6.44 $\pm$ 0.76	11.13 $\pm$ 1.19	19.39 $\pm$ 3.60	23.37 $\pm$ 1.18	63.18 $\pm$ 0.56	33.40 $\pm$ 1.21	33.49 $\pm$ 0.43
Neighbourhood $k = 50$	6.34 $\pm$ 0.56	10.01 $\pm$ 1.32	20.48 $\pm$ 1.00	22.64 $\pm$ 3.24	61.28 $\pm$ 2.19	32.64 $\pm$ 0.29	34.23 $\pm$ 0.77
Neighbourhood $k = 200$	6.50 $\pm$ 0.46	10.58 $\pm$ 2.83	20.26 $\pm$ 1.13	22.46 $\pm$ 1.23	61.24 $\pm$ 1.08	32.87 $\pm$ 0.76	34.42 $\pm$ 1.23
Distance: normalized L_2 (default)	6.59 $\pm$ 0.17	11.79 $\pm$ 1.34	20.48 $\pm$ 2.64	24.46 $\pm$ 2.68	61.69 $\pm$ 0.67	33.88 $\pm$ 1.05	35.24 $\pm$ 1.10
Distance: cosine (null control)	6.55 $\pm$ 0.15	10.50 $\pm$ 1.10	20.04 $\pm$ 2.64	23.91 $\pm$ 1.70	62.22 $\pm$ 1.29	33.23 $\pm$ 0.81	34.98 $\pm$ 0.51
Distance: raw L_2	6.92 $\pm$ 0.78	10.45 $\pm$ 3.00	18.95 $\pm$ 5.23	25.72 $\pm$ 2.31	62.81 $\pm$ 0.80	33.20 $\pm$ 0.89	35.08 $\pm$ 2.48
Strategy: mode window (default)	6.55 $\pm$ 0.15	10.18 $\pm$ 1.30	22.44 $\pm$ 4.59	23.28 $\pm$ 2.04	60.95 $\pm$ 3.01	33.05 $\pm$ 0.21	34.53 $\pm$ 0.92
Strategy: sparse band with FPS	6.60 $\pm$ 0.62	11.21 $\pm$ 1.26	21.13 $\pm$ 2.64	22.46 $\pm$ 0.16	63.34 $\pm$ 0.27	34.29 $\pm$ 0.30	34.30 $\pm$ 1.26
Strategy: sparse band, random	6.46 $\pm$ 0.48	11.06 $\pm$ 0.70	21.35 $\pm$ 4.82	22.92 $\pm$ 0.87	62.26 $\pm$ 0.89	32.33 $\pm$ 0.96	34.46 $\pm$ 1.08
Strategy: inverse-cluster	6.44 $\pm$ 0.75	10.66 $\pm$ 1.06	19.17 $\pm$ 2.47	21.65 $\pm$ 2.28	62.70 $\pm$ 0.45	33.50 $\pm$ 0.95	34.69 $\pm$ 1.10
Strategy: densest window	6.64 $\pm$ 0.46	11.90 $\pm$ 0.49	20.04 $\pm$ 1.00	24.28 $\pm$ 1.64	63.50 $\pm$ 0.60	33.80 $\pm$ 0.93	35.20 $\pm$ 0.80
Strategy: top tail, no diversity	6.39 $\pm$ 0.43	10.48 $\pm$ 1.03	19.83 $\pm$ 2.00	23.82 $\pm$ 1.50	62.86 $\pm$ 0.92	33.14 $\pm$ 1.02	34.66 $\pm$ 0.41

Table 22: Selector robustness on ImageNet-100-LT. Each row changes one component of the acquisition rule – candidate pool multiplier, upper cutoff, neighbourhood size, distance, or selection strategy – and holds the rest of the loop fixed. Mean $\pm$ standard deviation over three seeds. Cosine scoring rescales the normalized L_2 score by a positive constant and therefore selects the identical anchor set; that row is a null control, and its distance from the normalized L_2 row measures the end-to-end run-to-run spread of the pipeline. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
Candidate pool $\alpha = 1$	5.02 $\pm$ 0.14	8.62 $\pm$ 0.26	19.22 $\pm$ 0.71	14.92 $\pm$ 0.99	62.40 $\pm$ 0.12	31.47 $\pm$ 0.25	33.47 $\pm$ 1.05
Candidate pool $\alpha = 2$	5.08 $\pm$ 0.15	8.36 $\pm$ 0.22	19.07 $\pm$ 0.42	15.17 $\pm$ 0.27	62.81 $\pm$ 0.58	31.72 $\pm$ 0.27	33.19 $\pm$ 0.61
Candidate pool $\alpha = 4$ (default)	4.92 $\pm$ 0.15	8.48 $\pm$ 0.72	19.04 $\pm$ 0.03	14.60 $\pm$ 0.38	62.35 $\pm$ 0.54	31.84 $\pm$ 0.37	32.59 $\pm$ 0.49
Candidate pool $\alpha = 8$	4.85 $\pm$ 0.17	8.53 $\pm$ 0.41	19.17 $\pm$ 0.90	14.82 $\pm$ 0.62	62.15 $\pm$ 0.07	31.56 $\pm$ 0.23	33.58 $\pm$ 1.03
Upper cutoff $q = 0.95$	4.99 $\pm$ 0.11	8.57 $\pm$ 0.51	19.09 $\pm$ 0.51	14.68 $\pm$ 0.28	62.84 $\pm$ 0.45	31.88 $\pm$ 0.04	32.56 $\pm$ 0.45
Upper cutoff $q = 0.99$ (default)	4.95 $\pm$ 0.17	8.53 $\pm$ 0.69	19.40 $\pm$ 0.45	14.85 $\pm$ 0.49	62.29 $\pm$ 0.50	31.87 $\pm$ 0.16	32.44 $\pm$ 0.48
Upper cutoff $q = 0.995$	5.01 $\pm$ 0.07	8.34 $\pm$ 0.74	19.58 $\pm$ 0.83	14.87 $\pm$ 0.48	63.05 $\pm$ 0.47	32.06 $\pm$ 0.11	33.01 $\pm$ 0.34
Upper cutoff $q = 1.0$	4.99 $\pm$ 0.08	8.15 $\pm$ 0.61	18.52 $\pm$ 0.98	14.45 $\pm$ 0.19	62.84 $\pm$ 0.32	31.69 $\pm$ 0.30	32.99 $\pm$ 0.83
Neighbourhood $k = 10$	5.04 $\pm$ 0.09	8.99 $\pm$ 1.21	18.86 $\pm$ 0.35	14.90 $\pm$ 0.57	62.73 $\pm$ 0.39	31.69 $\pm$ 0.17	33.14 $\pm$ 0.40
Neighbourhood $k = 25$	4.97 $\pm$ 0.14	8.59 $\pm$ 0.21	19.05 $\pm$ 0.29	15.04 $\pm$ 0.85	62.68 $\pm$ 0.66	31.82 $\pm$ 0.16	33.56 $\pm$ 0.97
Neighbourhood $k = 50$	5.06 $\pm$ 0.17	8.62 $\pm$ 0.77	19.26 $\pm$ 0.67	14.92 $\pm$ 0.03	62.63 $\pm$ 0.40	31.71 $\pm$ 0.59	31.88 $\pm$ 0.57
Neighbourhood $k = 200$	5.17 $\pm$ 0.15	8.65 $\pm$ 0.59	19.22 $\pm$ 1.02	15.17 $\pm$ 0.10	62.80 $\pm$ 0.31	31.90 $\pm$ 0.13	33.91 $\pm$ 0.54
Distance: normalized L_2 (default)	4.97 $\pm$ 0.18	8.55 $\pm$ 0.67	19.42 $\pm$ 0.49	14.85 $\pm$ 0.50	62.27 $\pm$ 0.52	31.86 $\pm$ 0.16	32.60 $\pm$ 0.96
Distance: cosine (null control)	4.90 $\pm$ 0.13	8.52 $\pm$ 0.69	19.35 $\pm$ 0.54	14.75 $\pm$ 0.44	62.31 $\pm$ 0.53	31.78 $\pm$ 0.28	32.44 $\pm$ 0.83
Distance: raw L_2	4.85 $\pm$ 0.10	7.83 $\pm$ 0.39	19.13 $\pm$ 0.88	14.63 $\pm$ 0.34	62.81 $\pm$ 0.52	31.71 $\pm$ 0.04	32.28 $\pm$ 0.65
Strategy: mode window (default)	4.90 $\pm$ 0.12	8.52 $\pm$ 0.69	19.34 $\pm$ 0.54	14.75 $\pm$ 0.45	62.31 $\pm$ 0.52	31.78 $\pm$ 0.28	32.73 $\pm$ 0.75
Strategy: sparse band with FPS	5.19 $\pm$ 0.04	8.12 $\pm$ 0.39	19.53 $\pm$ 0.49	14.95 $\pm$ 0.75	62.43 $\pm$ 0.17	31.62 $\pm$ 0.32	33.38 $\pm$ 1.46
Strategy: sparse band, random	5.00 $\pm$ 0.13	8.70 $\pm$ 0.06	19.36 $\pm$ 0.88	14.95 $\pm$ 0.61	62.63 $\pm$ 0.19	31.64 $\pm$ 0.62	33.04 $\pm$ 0.82
Strategy: inverse-cluster	5.05 $\pm$ 0.02	8.83 $\pm$ 1.05	19.40 $\pm$ 0.81	15.06 $\pm$ 0.46	62.60 $\pm$ 0.50	31.99 $\pm$ 0.37	33.75 $\pm$ 0.44
Strategy: densest window	4.88 $\pm$ 0.10	8.44 $\pm$ 0.54	19.34 $\pm$ 0.59	14.95 $\pm$ 0.53	62.46 $\pm$ 0.12	31.62 $\pm$ 0.30	33.08 $\pm$ 0.64
Strategy: top tail, no diversity	4.90 $\pm$ 0.12	8.38 $\pm$ 0.29	19.11 $\pm$ 0.64	15.22 $\pm$ 0.03	62.05 $\pm$ 0.77	31.77 $\pm$ 0.13	32.41 $\pm$ 0.25

Table 23: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the k NN score distribution, holding every other part of the loop fixed. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
q0–25 (densest)	6.73 $\pm$ 0.36	9.63 $\pm$ 0.26	24.84 $\pm$ 5.10	23.37 $\pm$ 2.72	62.97 $\pm$ 0.82	33.96 $\pm$ 0.49	34.44 $\pm$ 0.62
q25–50	6.80 $\pm$ 0.39	10.28 $\pm$ 1.37	23.75 $\pm$ 2.10	22.74 $\pm$ 1.57	62.76 $\pm$ 0.64	33.20 $\pm$ 1.12	35.61 $\pm$ 0.23
q50–75	6.79 $\pm$ 0.63	10.24 $\pm$ 1.94	20.04 $\pm$ 4.64	23.55 $\pm$ 2.58	62.24 $\pm$ 0.37	32.80 $\pm$ 1.66	34.51 $\pm$ 1.27
q75–85	6.12 $\pm$ 0.33	9.74 $\pm$ 1.00	22.88 $\pm$ 4.58	23.46 $\pm$ 1.64	60.97 $\pm$ 1.05	31.90 $\pm$ 0.36	33.55 $\pm$ 0.46
q85–90	6.62 $\pm$ 0.39	9.95 $\pm$ 0.73	23.75 $\pm$ 3.09	25.45 $\pm$ 1.77	63.38 $\pm$ 0.78	33.47 $\pm$ 0.93	35.24 $\pm$ 0.69
q90–95	6.55 $\pm$ 0.55	10.46 $\pm$ 1.87	22.66 $\pm$ 3.94	25.27 $\pm$ 0.72	62.30 $\pm$ 1.15	33.49 $\pm$ 0.87	34.40 $\pm$ 0.62
q95–97	6.58 $\pm$ 0.20	9.55 $\pm$ 1.62	23.09 $\pm$ 1.00	23.55 $\pm$ 0.68	61.77 $\pm$ 2.13	32.75 $\pm$ 2.27	34.99 $\pm$ 1.19
q97–99	6.80 $\pm$ 0.17	11.38 $\pm$ 2.82	20.26 $\pm$ 3.98	26.27 $\pm$ 3.24	62.73 $\pm$ 1.49	33.13 $\pm$ 1.97	35.31 $\pm$ 1.55
q99–100 (extreme tail)	6.55 $\pm$ 0.28	10.78 $\pm$ 1.23	21.79 $\pm$ 0.75	23.82 $\pm$ 1.37	63.54 $\pm$ 0.92	33.65 $\pm$ 0.55	35.57 $\pm$ 0.91

Table 24: Repair utility by sparsity band on ImageNet-100-LT. Each row spends the identical acquisition budget on a different quantile range of the k NN score distribution, holding every other part of the loop fixed. Mean $\pm$ standard deviation over three seeds. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
q0–25 (densest)	4.97 $\pm$ 0.07	8.33 $\pm$ 0.72	20.00 $\pm$ 0.65	14.48 $\pm$ 0.27	62.04 $\pm$ 0.84	31.45 $\pm$ 0.05	31.88 $\pm$ 0.91
q25–50	5.16 $\pm$ 0.14	8.52 $\pm$ 0.37	19.58 $\pm$ 0.49	14.82 $\pm$ 0.73	62.95 $\pm$ 0.79	31.54 $\pm$ 0.09	32.88 $\pm$ 0.78
q50–75	5.06 $\pm$ 0.08	8.51 $\pm$ 0.44	19.63 $\pm$ 0.75	14.46 $\pm$ 0.81	62.41 $\pm$ 0.61	31.47 $\pm$ 0.56	32.61 $\pm$ 0.55
q75–85	4.97 $\pm$ 0.08	8.78 $\pm$ 0.45	19.54 $\pm$ 0.14	15.39 $\pm$ 0.20	62.52 $\pm$ 0.56	31.31 $\pm$ 0.43	32.23 $\pm$ 1.17
q85–90	5.04 $\pm$ 0.09	9.04 $\pm$ 0.48	19.61 $\pm$ 0.46	14.75 $\pm$ 0.65	62.57 $\pm$ 0.27	31.74 $\pm$ 0.34	32.82 $\pm$ 0.96
q90–95	4.92 $\pm$ 0.21	8.67 $\pm$ 0.75	19.60 $\pm$ 0.53	14.96 $\pm$ 0.46	62.40 $\pm$ 0.59	31.56 $\pm$ 0.54	32.48 $\pm$ 0.56
q95–97	4.88 $\pm$ 0.35	8.51 $\pm$ 1.09	19.40 $\pm$ 0.72	14.97 $\pm$ 0.49	62.32 $\pm$ 0.94	31.55 $\pm$ 0.43	31.63 $\pm$ 0.32
q97–99	5.02 $\pm$ 0.08	8.59 $\pm$ 0.43	19.63 $\pm$ 0.67	14.89 $\pm$ 0.56	62.64 $\pm$ 0.56	31.75 $\pm$ 0.61	32.59 $\pm$ 0.96
q99–100 (extreme tail)	5.17 $\pm$ 0.16	8.16 $\pm$ 0.23	19.10 $\pm$ 0.60	15.00 $\pm$ 0.53	62.81 $\pm$ 0.26	32.12 $\pm$ 0.43	32.15 $\pm$ 1.40

I ALGORITHM

Algorithm 1 BRIDGE training loop with mode-window selection

- 1: Initialize source set $D^{(0)}$, encoder $f_{\theta^{(0)}}$, frozen generator g_ϕ , stage lengths $\{E_c\}_{c=0}^{C-1}$
 - 2: **for** $c = 0, \dots, C - 1$ **do**
 - 3: Continue training $f_{\theta^{(c)}}$ on $D^{(c)}$ for E_c epochs with a fresh optimizer schedule
 - 4: **if** $c < C - 1$ **then**
 - 5: Compute normalized embeddings $Z^{(c)} = \{f_{\theta^{(c)}}(x_i) \mid x_i \in D^{(c)}\}$
 - 6: Compute mean k NN scores $\{d_i^{(c)}\}$ from $Z^{(c)}$
 - 7: Restrict to the 75th to 99th percentile band and locate the modal bin center $m^{(c)}$
 - 8: Form the candidate pool $C^{(c)}$ of size αB whose scores are closest to $m^{(c)}$
 - 9: Select $S^{(c)} \subseteq C^{(c)}$ with $|S^{(c)}| = B$ by greedy farthest-point sampling in embedding space
 - 10: Form $\tilde{D}^{(c)} = \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$ and set $D^{(c+1)} = D^{(c)} \cup \tilde{D}^{(c)}$
 - 11: **end if**
 - 12: **end for**
 - 13: Return the final encoder $f_{\theta^{(C-1)}}$
-

Table 25: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	7.15 $\pm$ 0.19	10.92 $\pm$ 0.40	32.90 $\pm$ 2.72	25.00 $\pm$ 1.25	60.46 $\pm$ 0.46	34.96 $\pm$ 0.20	41.95 $\pm$ 1.43
SimCLR, with BRIDGE	8.18 $\pm$ 0.42	11.82 $\pm$ 0.43	36.60 $\pm$ 2.85	29.80 $\pm$ 2.98	60.85 $\pm$ 0.35	36.33 $\pm$ 0.14	43.93 $\pm$ 1.09
MoCo v3, SSL baseline	7.57 $\pm$ 0.23	11.05 $\pm$ 1.31	30.28 $\pm$ 1.36	29.26 $\pm$ 3.13	59.96 $\pm$ 0.39	34.47 $\pm$ 0.01	44.79 $\pm$ 1.17
MoCo v3, with BRIDGE	9.55 $\pm$ 0.34	13.42 $\pm$ 0.98	36.82 $\pm$ 6.07	31.25 $\pm$ 4.00	62.36 $\pm$ 0.46	36.30 $\pm$ 0.13	47.55 $\pm$ 0.86
DINO, SSL baseline	8.10 $\pm$ 0.25	11.40 $\pm$ 1.03	38.13 $\pm$ 5.24	27.54 $\pm$ 2.34	58.93 $\pm$ 0.94	33.56 $\pm$ 1.75	44.63 $\pm$ 1.47
DINO, with BRIDGE	9.49 $\pm$ 0.88	13.28 $\pm$ 0.58	37.25 $\pm$ 2.61	34.60 $\pm$ 3.62	61.30 $\pm$ 0.07	35.23 $\pm$ 0.31	46.78 $\pm$ 0.77
MAE, SSL baseline	1.54 $\pm$ 0.19	2.44 $\pm$ 1.00	10.46 $\pm$ 5.88	10.05 $\pm$ 3.03	37.69 $\pm$ 3.73	13.20 $\pm$ 1.94	19.93 $\pm$ 2.13
MAE, with BRIDGE	1.56 $\pm$ 0.43	3.18 $\pm$ 0.70	8.50 $\pm$ 2.26	8.61 $\pm$ 2.20	41.46 $\pm$ 2.20	15.00 $\pm$ 2.40	22.19 $\pm$ 2.35

Table 26: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both arms of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean $\pm$ standard deviation over three seeds. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	6.26 $\pm$ 0.17	7.09 $\pm$ 0.17	36.53 $\pm$ 0.89	19.16 $\pm$ 0.64	66.05 $\pm$ 0.48	36.72 $\pm$ 0.36	41.21 $\pm$ 0.72
SimCLR, with BRIDGE	6.52 $\pm$ 0.22	7.56 $\pm$ 0.23	38.29 $\pm$ 0.83	19.93 $\pm$ 0.53	65.70 $\pm$ 0.85	36.46 $\pm$ 0.33	42.91 $\pm$ 0.78
MoCo v3, SSL baseline	6.65 $\pm$ 0.38	7.71 $\pm$ 0.16	38.91 $\pm$ 1.03	23.56 $\pm$ 1.12	69.29 $\pm$ 0.25	39.98 $\pm$ 0.44	45.14 $\pm$ 0.46
MoCo v3, with BRIDGE	6.74 $\pm$ 0.26	8.21 $\pm$ 0.18	40.51 $\pm$ 1.45	24.91 $\pm$ 0.45	65.41 $\pm$ 0.21	36.24 $\pm$ 0.46	47.71 $\pm$ 1.53
DINO, SSL baseline	6.05 $\pm$ 0.27	7.73 $\pm$ 0.52	38.52 $\pm$ 1.19	19.26 $\pm$ 0.16	62.69 $\pm$ 2.42	33.98 $\pm$ 2.80	45.28 $\pm$ 0.98
DINO, with BRIDGE	6.39 $\pm$ 0.22	8.16 $\pm$ 0.26	40.72 $\pm$ 0.31	21.79 $\pm$ 1.15	64.38 $\pm$ 3.51	35.59 $\pm$ 2.51	47.14 $\pm$ 0.29
MAE, SSL baseline	1.00 $\pm$ 0.14	2.24 $\pm$ 0.29	5.30 $\pm$ 0.52	4.58 $\pm$ 0.17	31.34 $\pm$ 1.65	10.96 $\pm$ 0.44	11.75 $\pm$ 0.38
MAE, with BRIDGE	1.38 $\pm$ 0.37	2.50 $\pm$ 0.70	7.54 $\pm$ 2.13	5.35 $\pm$ 0.76	35.58 $\pm$ 7.14	13.44 $\pm$ 3.46	15.46 $\pm$ 2.68